



HARVEST in 5 Steps: A Simple Guide for Agricultural Professionals

1. What is HARVEST?

HARVEST is a user-friendly digital agriculture platform developed through the ICICLE AI Institute. It helps farmers, extension agents, and agribusiness professionals make smarter and faster field decisions using artificial intelligence (AI).

Think of it as a bridge between your field data and actionable insights showing you where to focus attention, e.g. how to manage weeds, and how to make better crop decisions.

 HARVEST

Home **Dashboard** Help Contact Us

Dashboard

OBJECT DETECTION

OBJECT CLASSIFICATION

Job Composer

Please check components to compose a new

Preprocessing Data Collection ☐	Data watch User Data Pipeline Watch ☐	Data Movement Extract Frames ☐	Preprocessing Data Preprocessing ☐
Preprocessing Crop Images ☐	Training Model Training ☐	Inference Model Inference ☐	Visualization Visualize Results ☐
Detection Object Detection ☐	Classification Object Classification ☐	Shapefile Shapefile Generation ☐	

2. Why HARVEST Matters

- **For Farmers:** The tool can save time, reduce chemical waste, and help target field issues precisely.
- **For Extension Agents:** Provides digital tools to support farmer training and field decision-making.
- **For Agribusiness Professionals:** Demonstrates how AI can improve product placement, precision agriculture, and decision-support systems.

3. The HARVEST Process in 5 Simple Steps

Step 1: Data Collection – “It All Starts in the Field”

- A drone flies over the farm to capture high-resolution images of crops/field.
- These images cover the entire plot with pinpoint accuracy.
- Farmers or service providers can use standard agricultural drones for this step.

Benefit: You get complete and reliable images of your farm.

Step 2: Data Upload and Preparation – “Getting Your Images Ready for Analysis”

- The captured images are uploaded to HARVEST’s secure cloud system.
- The system processes each image - aligning and preparing them for analysis.
- Every image is tagged with location details, ensuring that analysis matches your actual field layout.

Benefit: HARVEST organizes your images and prepares them for accurate insights.

Step 3: Model Training - “Teaching the System What to Look For”

- The user (or researcher) selects what the model should learn - for example, identifying weed types, crop stress, or growth stages.
- The system uses high-performance computers to train an AI model based on those examples.
- Once training is done, the model is ready to make predictions for new fields or seasons.

Benefit: HARVEST learns from your data, which means the more it’s used, the smarter it gets.

Step 4: Inference - “Turning Data into Action”

- New images from the same or another field are uploaded.
- The trained model scans the images, detecting things like weeds or crop stress automatically.
- Results are generated quickly and shown as maps or tables that summarize what’s happening in the field.

Benefit: You see where problems exist without checking the entire field manually.

Step 5: Visualization and Action - “See It, Understand It, Act”

- HARVEST creates heatmaps that show crop conditions.
- It also generates shape files (field maps) that mark exactly where actions like spraying should take place.
- These maps can be used directly in modern precision sprayers or shared with field teams for on-site decisions.

Benefit: Enables precision agriculture, treating only where needed, saving time and money while protecting the environment.

4. What Makes HARVEST Different

- **End-to-End Simplicity:** From data capture to decision all in one platform.
- **Model Accuracy:** Learns and improves with every dataset.
- **Scalable:** Works for small farms, research plots, and large agribusiness operations.

5. How You Can Be Part of It

- Watch the HARVEST demo videos to see action
- Complete the feedback survey, share what worked, what didn't, and what could be improved.

Your feedback will help improve future versions of HARVEST and shape digital tools that truly work for agricultural communities.

In summary, HARVEST can help you make informed, data-driven decisions from identifying field problems to applying precise solutions.

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